

CEREBRO-X

Computational Drug-DDS Engineering Report

Field	Value
Drug Name	Nusinersen
Molecular Class	oligonucleotide
Molecular Weight	1593.8 Da
Recommended DDS	ApoE-LNP
Carrier Type	Inp
Surface Ligand	ApoE
BBB Enhancement	N/A%
Composite Score	100.0/100
Disease Indication	CNS
Report Generated	2026-05-09 18:37 UTC
Created by	Muhammad Talaat CEREBRO-X
Modules Completed	50/62 science modules

■ This report is generated from computational models. All predictions must be validated experimentally before clinical application.

1. Drug Molecular Profile

Property	Value	Source / Method
Name	Nusinersen	Input
MW (Da)	1593.8	ChEMBL / UniProt
LogP	0.18	ADMET predictor
Half-life (days)	N/A	PubMed NLP
Protein Binding (%)	N/A	Calculated
BBB Penetration (%)	96.86	Pardridge 2012
Molecule Class	oligonucleotide	Input / inferred
Aqueous Solubility	N/A	ADMET
P-gp Substrate	Unknown	QSAR panel
ADMET Score	N/A	Multi-endpoint
ADMET Grade	N/A	A-F scale

1a. Drug Delivery Barriers Identified (Auto-detected)

The pipeline automatically identified 1 delivery barriers based on molecular data. Each is resolved by the recommended DDS.

[CRITICAL] Large molecular weight -- passive BBB diffusion impossible	
Evidence:	MW = 1.6 kDa (Lipinski Rule 5 limit: 500 Da for passive)
Without DDS:	0% brain penetration -- purely physical barrier
With DDS:	Active transcytosis delivers 10.0% payload to CNS
DDS Solution:	Inp encapsulation (95% EE)
Scientific basis:	BBB tight junctions (pore ~1nm) physically block molecules >500Da. Encapsulation in Inp (80nm) enables receptor

2. DDS Ranking — Composite Score Methodology

All DDS formulations ranked by Composite Score = weighted combination of DLVO colloidal stability, transcytosis driving force, endosomal escape, stealth index, BBB receptor affinity, and CNS bioavailability. Scores are physics-derived — not estimated.

DDS Name	Carrier	Ligand	Score	DLVO kT
ApoE-LNP	lnp	ApoE	100.00	-6.24
RVG29-Liposome-pH	liposome	RVG29	100.00	14.84
Tf-PEG-Liposome	liposome	Transferrin	100.00	8.90
RVG29-PLGA-NP	plga	RVG29	100.00	5.83
Tf-SLN-Stealth	solid_lipid	Transferrin	100.00	-0.12
Lf-Polymer-Micelle	micelle	Lactoferrin	97.00	-4.06
Bare-PLGA	plga	None	78.00	-16.16
AAV9-vector	aav9	None	53.00	-2.37

2a. Recommended DDS — Detailed Profile

Parameter	Value	Biophysical Meaning
Composite Score	100.0	Overall Drug+DDS suitability (0-100)
BBB Enhancement	N/A%	% of dose crossing BBB via receptor transcytosis
CNS Bioavailability	N/A%	% dose reaching brain as free drug
DLVO Stability	-6.2 kT	>25kT = colloiddally stable in blood
Endosomal Escape	N/A	Fraction escaping lysosomes (0-1)
Stealth Index	N/A	MPS evasion (0=opsonised, 1=invisible)
Protein Corona	N/A nm	Corona thickness; thicker = faster clearance
MPS Clearance	N/A h	Hours before liver/spleen removes DDS
Payload Efficiency	N/A%	% dose delivered as active drug to CNS
CARPA Risk	0.20	Complement activation risk (0=safe)

3. PBPK-CNS Digital Twin — 6-Compartment Simulation

Physiologically Based Pharmacokinetic model solved with scipy Radau ODE integrator (stiff system). Compartments: Plasma, BBB endothelium, Brain ISF, Brain cells, CSF, Peripheral. All parameters from molecular data — not estimated. Reference: Pardridge 2012; Bhatt 2013.

PK Metric	Value	Compartment	Source
Model	6-compartment ODE (Radau stiff-solver) Pardridge 2012		CEREBRO-X PBPK
Cmax (brain)	5835.415200 µg/mL	Brain/CNS	Peak brain concentration
AUC (brain)	82811.95050 µg·h/mL	Brain/CNS	Total brain exposure
AUC (plasma)	30708.270 µg·h/mL	Plasma	Systemic exposure
Kp,brain	2.69673	Brain/Plasma	Partition coefficient
BBB penetration	0.0000%	BBB	
Kp,uu brain	5.40289	Unbound	Unbound Kp (pharmacodynamically relevant)
t_ above 10%Cmax	33.0 h	Brain ISF	Duration above therapeutic threshold
Glymphatic t1/2	72.0 h	Brain	Clearance half-life in brain
BBB PS_in	587.860 mL/h	BBB	Permeability-surface area product
BBB PS_out	251.310 mL/h	BBB	Efflux transport rate
Disease state	alzheimer_2	BBB	BBB integrity modifier applied

3a. PBPK Time-Course (sampled points)

t (h)	Plasma (µg/mL)	Brain ISF (µg/mL)	Ratio (Kp)
0.0	23.7500	0.00000	0.00000
6.5	23.3356	60.02434	2.57222
13.0	35.5395	94.99681	2.67300
19.5	55.3607	148.88824	2.68942
26.0	86.7962	234.02378	2.69625
32.5	136.4753	368.37848	2.69923
39.3	218.5177	590.11919	2.70056
45.8	344.2554	929.86425	2.70109
52.3	542.4910	1465.43626	2.70131
58.8	854.9861	2309.65878	2.70140
65.3	1347.5712	3640.37293	2.70143
72.0	2160.1094	5835.41520	2.70144

4. In-Silico Drug Release Profile

Release model: First Order | Release order: Zero-order (constant rate) | Max EE: 95%

Metric	Blood (pH 7.4)	Endosomal (pH 5.5)	Significance
t50 (50% release)	2.9 h	2.9 h	Faster endo = better escape
t90 (90% release)	9.2 h	N/A	Complete release window
Rate constant k	0.2500 /h	0.2500 /h	Higher endo = pH-responsive
Max released	95%	95%	= encapsulation efficiency

5. Shelf-Life & Degradation Predictor

Grade: MARGINAL (3-6 months) | t90 = 98 days | Dominant pathway: Aggregation | Storage: 4 degC

Pathway	Rate (k, /day)	Relative %
Hydrolysis	0.000061	6%
Oxidation	0.000160	15%
Aggregation	0.000781	73%
Drug leakage	0.000074	7%

6. Colloidal Stability — DLVO Theory + Protein Corona

DLVO potential energy $V_{\text{total}} = -6.2 \text{ kT}$ (UNSTABLE $\times$ — REFORMULATE) | Threshold: $>25 \text{ kT}$ prevents aggregation in blood. Reference: Verwey & Overbeek 1948; Derjaguin 1987.

Parameter	Value	Equation	Significance
DLVO V_{total} (kT)	-6.2	$V_{\text{vdW}} + V_{\text{EDL}}$	$>25\text{kT} = \text{stable}$
V_{vdW} (attraction)	N/A	$-A \cdot R / (12h)$	Hamaker constant
V_{EDL} (repulsion)	N/A	$64\pi \epsilon R \gamma^2 e^{-\kappa h}$	Zeta potential
Debye length (nm)	N/A	$1/\kappa$	Ionic screening
Protein corona (nm)	N/A	Langmuir model	Stealth reduction
Opsonin index	N/A	IgG + C3b binding	$<0.3 = \text{good}$
Stealth index	N/A	PEG brush model	$0.6+ = \text{good}$
MPS clearance (h)	N/A	Michaelis-Menten	$>12\text{h}$ adequate

7. Nanotoxicity & Immunogenicity Screening

Immunogenicity grade: **CAUTION** | Score: 37/100

Test	Score / Result	Risk	Threshold
CARPA (complement)	0.660	HIGH	$>0.6 = \text{HIGH}$
Anti-PEG antibody	0.075	LOW	$0.5+ = \text{caution}$
MPS uptake	0.487	—	$>0.6 = \text{rapid clearance}$
Cytokine storm	—	LOW	Charge-dependent
Platelet activation	—	LOW	Cationic $>200\text{nm}$

Mitigations:

→ Consider pre-medication with antihistamines (H1/H2 blockers)

8. Off-Target Toxicity — 50-Receptor QSAR Panel

Overall: SAFE | Cardiac risk: NO ✓ | Hepatic risk: NO ✓ | CNS off-target: NO ✓

Receptor	Free Drug Score	In-DDS Score	Risk
hERG_K+	0.000	0.000	LOW
Nav1.5	0.000	0.000	LOW
Cav1.2	0.000	0.000	LOW
CYP3A4_inhib	0.134	0.070	LOW
CYP2D6_inhib	0.196	0.103	LOW
CYP2C9_inhib	0.098	0.052	LOW
CYP1A2_inhib	0.257	0.135	LOW
CYP2C8_inhib	0.125	0.066	LOW
OATP1B1	0.217	0.114	LOW
MDR1_Pgp	0.000	0.000	LOW
BCRP	0.128	0.067	LOW
MRP2	0.155	0.081	LOW
DAT_dopamine	0.000	0.000	LOW
SERT_serotonin	0.000	0.000	LOW
5HT2A	0.000	0.000	LOW
D2_dopamine	0.000	0.000	LOW
GABA_A	0.000	0.000	LOW
NMDA	0.000	0.000	LOW
sigma1	0.225	0.118	LOW
ERalpha_estrogen	0.000	0.000	LOW

9. Advanced Science Modules — Key Outputs

9a. Competitive DDS Landscape (ClinicalTrials.gov)

Our DDS: BBB Score = 100 | Position: SUPERIOR | Active trials found: 3

NCT ID	Title	Phase	N
NCT04899908	Stereotactic Brain-directed Radiation With or	[PHASE2]	134
NCT06855368	Biomarkers for Monitoring Dementia Progressio	[?]	300
NCT06846658	Exploring the Olfactory Mucosa, Blood and Uri	[?]	180

9b. Patient Subgroup Stratification

Overall response: 28.2% | Best: Very elderly (>80) (32%) | Worst: Young adults (18-40) (24%) | Recommendation:
Highest efficacy in Very elderly (>80) (predicted response 32%). Dose reduce by

Subgroup	Pop. Freq.	CNS Bioavail	Response%	Tox Risk	Dose Adj.
Young adults (18-40)	20% of patients	7.1%	24.5%	LOW	x1.29 standard dose
Middle-aged (40-65)	35% of patients	8.1%	28.3%	LOW	x1.14 standard dose
Elderly (65-80)	30% of patients	8.3%	29.0%	LOW	x0.72 standard dose
Very elderly (>80)	15% of patients	8.9%	31.5%	MODERATE	x0.46 standard dose

9d. Lysosomal Trafficking Predictor

Cytosolic route: 34% | Lysosomal degradation: 64% | Nuclear entry: 2% | Concern: HIGH -- significant lysosomal degradation expected | Mitigation: Add proton sponge polymer (PEI or PAMAM) to improve endosomal escape

10. Synthetic Clinical Trial — N=500 Virtual Patients

Monte Carlo simulation of Phase 1 trial. Patient covariates: age, weight, CYP3A4 genotype, renal function, BBB integrity by age. Pharmacokinetics computed per-patient from PBPK model. Reference: FDA PBPK guidance 2018; Lalonde 2007.

NO-GO / REFORMULATE Go/No-Go	26.0% Response Rate	0.0% Severe AE	1.07 mg/kg Optimal Dose	500 N Patients
--	-----------------------------------	------------------------------	---------------------------------------	------------------------------

Endpoint	Result	Benchmark	Status
Overall response rate	26.0%	>60% = GO	✗
Severe AE rate	0.0%	<5% = GO	✓
Young adults (<65)	29.4%	—	—
Elderly (>65)	24.9%	—	—
Mild AE	0.4%	—	—
Renal AE	0.0%	<15%	✓

Recommendation: Predicted Phase 1 success rate: 26%. Dose 1.07 mg/kg recommended. Monitor renal function in 0% of elderly patients.

11. Manufacturing & Sterilization Modules

11a. Terminal Sterilization Survivability

Recommended method: Gamma 25kGy | Cost implication: Gamma or E-beam (~3x cost) | Feasible methods: gamma 25kGy, aseptic filtration 022um

Method	Survives?	Detail
autoclave 121C	NO ✗	Biologic MW=2kDa -- THERMAL DENATURATION
gamma 25kGy	YES ✓	Survives gamma sterilization
aseptic filtration 022um	YES ✓	Size 80 nm < 220 nm -- filterable
VHP H2O2	NO ✗	Surface ligands may be oxidized -- validate targeting post-V

11b. Lyophilization Cycle Optimizer

Parameter	Value	Standard
Tg' (cryo.	-30°C	Formulation-specific
Primary drying T	-32.0°C	Must be < Tg'
Primary P	0.1478 mbar	0.5×P_ice
Primary time	25.6 h	Pikal 2002 model
Secondary T	25.0°C	+25°C
Cycle total	39.6 h	Include ramp time
Cake collapse	OK: Tg within safe margin	Must say OK

11c. Cryo-Chain Excursion Predictor

Excursion: -20.0°C for 4.0h | EE: 95.0% → 94.0% (loss: 1.0%) | Decision: RELEASE BATCH | Confidence: 98%

11d. Adversarial Stress-Testing

Grade: 3/5 scenarios passed -- CONDITIONAL | 3/5 scenarios passed.

Scenario	Pass/Fail	Detail
Fever 40C	FAIL ✗	Tm=40.0degC -- RISK: may partially melt at fever temperature
Shear Stress	PASS ✓	Carrier withstands capillary shear
Oxidative Stress	PASS ✓	Polymeric carrier -- relatively resistant to oxidative stress
High Protein Plasma	FAIL ✗	Stealth index 0.50: RISK: rapid clearance in protein-rich plasma
Repeated Dosing	PASS ✓	PEG density outside typical ABC range -- lower re-dosing risk

12. Regulatory Compliance & Supply Chain

12a. FDA 21 CFR Part 11 Compliance

Status: 21 CFR Part 11 COMPLIANT -- electronic records with audit trail | Audit entries: 0 | Data integrity: SHA-256 hash chain -- tamper-evident

12b. Geopolitical Supply-Chain Risk Assessment

Overall supply risk: HIGH | Supply chain score: 0/100 | Recommendation: Establish 6-month buffer stock for high-risk materials

Material	Risk	Source	HHI Index	Mitigation
ApoE3 peptide	VERY HIGH	Custom synthesis	0.85	Qualify alternative supplier

12c. Digital Pharmacovigilance — Post-Elimination Fate

Unchanged excretion: 19.1% | Eco-persistence: 0.5 days | Environmental risk: LOW | Renal dose adjustment: Reduce by 50%

13. Literature Citations — PubMed Auto-Mined

Top relevant publications retrieved live from PubMed E-utilities API based on drug name + carrier type. These papers confirm or contextualize the computational predictions above.

[1] Pardridge WM (2012). Drug transport across the BBB. JCBFM. PMID:33567412

[2] Kreuter J (1994). Colloidal drug delivery. J Microencapsul. PMID:31270248

14. Biophysical Binding & CNS-Specific Modules

14a. FEP+ Binding Affinity (LIE approximation)

Ligand: apoe | $\Delta G_{\text{single}} = -5.00$ kcal/mol | $\Delta G_{\text{avidity}} = -2.59$ kcal/mol | $K_d = 14975652$ nM (Weak (>100nM)) | $n_{\text{ligands}} = 16085$ | Residence time = 66.8 s

14b. Glymphatic Clearance Simulation

ECM binding index = 0.522 | $t_{1/2}$ waking = 9.5 h | t_{90} clearance = 31.4 h | CAUTION: small particles cleared rapidly by glymphatics

14c. Microglial Activation & Neuroinflammation

Neuroinflammation score = 0.055 | Risk: LOW -- safe for CNS application | IL-6 fold-change = 1.6× | $\text{TNF-}\alpha = 1.4\times$
→ No mitigation needed

14d. FUS-Responsive Nanocarrier Design

Frequency = 0.5 MHz | MI target = 0.40 | BBB open window = 42 min | FUS uptake = 68.0% | Inject 80 nm DDS 5 min before FUS. FUS opens BBB for 42 min. Predicted 68% enhan

14e. 62-Principle C+ Flow — Class A Surrogate (Top-1 DDS)

Top-1 DDS: ApoE-LNP **Composite:** 76.2/100 **Verdict:** GOOD

ApoE-LNP: composite CNS-principle score = 76.2/100 (GOOD). Carrier: lnp, size 0.0 nm, ζ +0.0 mV, ligand: ApoE. Strongest group: G4 Safety (97.0/100); weakest: G8 Translational (0.0/100). Evaluated against all 57 Class A surrogate principles.

CNS Principle Group Rollups

CNS Group	Score (0-100)
G1 CNS Delivery	51.0
G2 Release Kinetics	67.6
G3 Stability	63.3
G4 Safety	97.0
G5 Glymphatic BBB	60.0
G6 Manufacturability	66.9
G7 DrugDDS Fit	63.4
G8 Translational	0.0

Top Strengths (Class A Surrogate)

Principle	Score	Method	Reference
P03	100.0	Novelty = 100 - frequency_of_combo_in_CNS_trials,	ClinicalTrials.gov landscape
P05	100.0	% subgroups responding (lower CYP risk = more univ	Whirl-Carrillo M et al (2012) Clin
P09	100.0	100 - sum(metabolite organ accumulation risks)	Djoumbou-Feunang Y et al (2019) J C

Weak Spots (Below 60)

Principle	Score	Method	Reference
P60	15.0	Swarm capability (most carriers: no)	Servant A et al (2015) Sci Robot 1:
P44	1.0	AUC over 24h therapeutic window proxy	Bies RR et al (2019) Annu Rev Pharm
P13	0.7	AUC_brain/AUC_plasma proxy; 3-compartment surroga	Hammarlund-Udenaes M et al (2008) P

14e-ii. Complete 62-Principle Scoreboard (P01 → P62)

All 62 principles in canonical order, covering Class A (Surrogate), Class B (Deep Physics), and Class C (Translational).

P#	Cls	Title	Score	Conf.	Method
P01	A	Adversarial Stress-Testing Engine	57.1	LOW	Full MD stress simulation — surrogate value confir
P02	A	Cross-Species PK Scaling (Allometric)	75.0	MODERATE	Multi-parameter allometric scaling (Mahmood 2007):
P03	A	Competitive DDS Landscape Radar	0.0	—	
P04	A	Quantum Coherence Transport Model	0.2	LOW	Full QM tunneling (QCElemental + ASE) — surrogate
P05	A	In-Silico Patient Subgroup Stratifier	0.0	—	
P06	A	Lysosomal Trafficking Predictor	0.0	—	
P07	A	Real-Time Literature Mining (PubMed)	0.0	—	
P08	A	Degradation Kinetics Under Oxidative S	45.0	LOW	Full radical-chain reaction MD — surrogate value c
P09	A	Digital Pharmacovigilance Engine	0.0	—	
P10	A	LNP Ionization State Predictor	100.0	MODERATE	Constant-pH MD simulation — surrogate value confir
P11	A	Formulation Instability Fingerprint	100.0	MODERATE	DFT bond dissociation energies — surrogate value c
P12	A	CNS Disease-Stage-Aware Dosing	95.8	MODERATE	Stage-specific PBPK with full BBB physiology — sur
P13	A	PBPK Digital Twin	4.2	MODERATE	3-compartment PBPK ODE (scipy.integrate.odeint): b
P14	A	In-silico Dissolution & Release Profil	0.0	—	
P15	A	Shelf-life & Degradation Predictor	0.0	—	
P16	A	Scale-up & Manufacturability	50.0	LOW	CFD of 1000L bioreactor — surrogate value confirme
P17	A	Nanotoxicity & Immunogenicity Screenin	0.0	—	
P18	A	Active Targeting & Receptor Binding	92.0	HIGH	MM/GBSA-style ΔG estimate from validated ligand-re
P19	A	QbD - Quality by Design Engine	0.0	—	
P20	A	Cost-Efficiency Engine	0.0	—	
P21	C	Pre-IND Regulatory Reports	0.0	—	skipped_deep_validation_insufficient
P22	A	Protein Corona Predictor	0.0	—	
P23	A	Dynamic Crystal Polymorphism	94.0	MODERATE	cCSP via DFT — surrogate value confirmed (full-phys
P24	A	Shear-Stress & Scale-Up Collapse	60.0	MODERATE	Full CFD + MD coupling — surrogate value confirmed
P25	A	Extractables & Leachables	0.0	—	
P26	A	Microbiome-Excipient Interactions	0.0	—	
P27	A	Lyophilization Cycle Optimizer	0.0	—	
P28	A	3D-Printed Polypill Rheology	0.0	—	
P29	A	Biomimetic & Exosome Engineering	65.0	MODERATE	Full membrane-fusion MD — surrogate value confirme
P30	A	QM/MM Stimuli-Responsive Cleavage	45.0	LOW	Full QM/MM (PySCF or ORCA) — surrogate value confi
P31	A	In-Silico Biodistribution	40.0	MODERATE	7-organ whole-body distribution with size + charge
P32	C	Automated FTO & IP Evader	0.0	—	skipped_deep_validation_insufficient
P33	A	BBB Quantum Breaker (Trojan-Horse Desi	54.6	LOW	Atomistic docking + SMD pulling — surrogate value

P#	Cls	Title	Score	Conf.	Method
P34	A	DNA Logic Gates & Bio-computing	0.0	—	
P35	A	Microgravity Formulation Engine	0.0	—	
P36	A	Geopolitical Supply-Chain Resilience	0.0	—	
P37	A	Eco-Destructible Pharma	0.0	—	
P38	A	Glymphatic Clearance Trap	96.1	HIGH	Stokes-Einstein $D = k_B \cdot T / (6\pi\eta r)$ at 37°C, $\eta_{CSF} =$
P39	A	Microglial Activation & Neuroinflammat	0.0	—	
P40	A	Intranasal-to-Brain Delivery	55.0	LOW	Full nasal cavity CFD — surrogate value confirmed
P41	A	Exosome Cargo Loading Thermodynamics	40.0	LOW	Full membrane mechanics MD — surrogate value confi
P42	A	Region-Specific Spatiotemporal Navigat	0.0	—	
P43	A	FUS-Responsive Nanocarriers	25.0	LOW	Full acoustic radiation force MD — surrogate value
P44	A	CNS-Specific PBPK Time-Machine	0.0	MODERATE	2-compartment CNS-PBPK ODE with glymphatic clearan
P45	C	FDA 21 CFR Part 11 Compliance	0.0	—	skipped_deep_validation_insufficient
P46	A	Polypharmacy & DDI Simulator	0.0	—	
P47	B	Free Energy Perturbation (FEP+)	44.5	MODERATE	Enhanced FEP+: Vina baseline + hydration correctio
P48	A	Off-Target Toxicity & QSAR (50-recepto	0.0	—	
P49	A	Organ-on-a-Chip Simulator	0.0	—	
P50	A	Cryo-Chain Thermal Excursion Predictor	71.5	MODERATE	Full coarse-grained MD lipid phase transition — su
P51	A	Terminal Sterilization Survivability	60.0	MODERATE	Full radical-chain damage MD — surrogate value con
P52	A	Continuous Manufacturing Digital Twin	0.0	—	
P53	A	Dark Data & Negative Results Vault	0.0	—	
P54	A	Pharmacogenomic-Guided Targeting	0.0	—	
P55	C	Automated Grant & NIH Proposal Generat	0.0	—	skipped_deep_validation_insufficient
P56	C	Patentability Score Engine	0.0	—	skipped_deep_validation_insufficient
P57	A	Microfluidics & LNP Synthesis Digital	50.0	LOW	Full CFD of mixer geometry — surrogate value confi
P58	A	Impurity Cascade Predictor	80.0	MODERATE	Full QM/MM cascade simulation — surrogate value co
P59	A	4D Shape-Shifting Carriers	30.0	LOW	Full coarse-grained MD morphological transition —
P60	A	Swarm Nanorobotics Intelligence	0.0	—	
P61	A	Synthetic Clinical Trials & Virtual Hu	30.1	LOW	Full Monte-Carlo population PBPK — surrogate value
P62	A	Biobetter / Supergeneric Generator	0.0	—	

62 principles evaluated. Composite: 76.2/100 (GOOD)

14f. Class B Deep Physics Validation (Top-1 DDS)

Verdict: **FAILED** **Pass rate:** 42.9% (12/28 principles validated, threshold 70%)

Deep validation: 12/28 principles passed (42.9%). Verdict: FAILED.

Per-principle deep validation

P#	Validated	Score	Value	Conf.	Method
P01	✗	57.1	57.15	LOW	Full MD stress simulation — surrogate value confirmed (
P02	✓	75.0	47.05	MODERATE	Multi-parameter allometric scaling (Mahmood 2007): half
P04	✗	0.2	0.0009	LOW	Full QM tunneling (QCElemental + ASE) — surrogate value
P08	✗	45.0	45.0	LOW	Full radical-chain reaction MD — surrogate value confir
P10	✓	100.0	100	MODERATE	Constant-pH MD simulation — surrogate value confirmed (
P11	✓	100.0	100	MODERATE	DFT bond dissociation energies — surrogate value confir
P12	✓	95.8	95.75	MODERATE	Stage-specific PBPK with full BBB physiology — surrogat
P13	✗	4.2	0.0021	MODERATE	3-compartment PBPK ODE (scipy.integrate.odeint): blood
P16	✗	50.0	50.0	LOW	CFD of 1000L bioreactor — surrogate value confirmed (fu
P18	✓	92.0	-9.2	HIGH	MM/GBSA-style ΔG estimate from validated ligand-recepto
P23	✓	94.0	94.0	MODERATE	CSP via DFT — surrogate value confirmed (full-physics H
P24	✓	60.0	60.0	MODERATE	Full CFD + MD coupling — surrogate value confirmed (ful
P29	✓	65.0	65	MODERATE	Full membrane-fusion MD — surrogate value confirmed (fu
P30	✗	45.0	45.0	LOW	Full QM/MM (PySCF or ORCA) — surrogate value confirmed
P31	✗	40.0	0.02	MODERATE	7-organ whole-body distribution with size + charge + li
P33	✗	54.6	54.6	LOW	Atomistic docking + SMD pulling — surrogate value confi
P38	✓	96.1	9.6	HIGH	Stokes-Einstein $D = k_B \cdot T / (6\pi\eta r)$ at 37°C, $\eta_{CSF} = 7 \times 10$ ■
P40	✗	55.0	55	LOW	Full nasal cavity CFD — surrogate value confirmed (full
P41	✗	40.0	40	LOW	Full membrane mechanics MD — surrogate value confirmed
P43	✗	25.0	25	LOW	Full acoustic radiation force MD — surrogate value conf
P44	✗	0.0	0.0	MODERATE	4-compartment CNS-PBPK ODE with glymphatic clearance: B
P47	✗	44.5	-4.05	MODERATE	Enhanced FEP+: Vina baseline + hydration correction (Lo
P50	✓	71.5	71.5	MODERATE	Full coarse-grained MD lipid phase transition — surroga
P51	✓	60.0	60	MODERATE	Full radical-chain damage MD — surrogate value confirme
P57	✗	50.0	50	LOW	Full CFD of mixer geometry — surrogate value confirmed
P58	✓	80.0	80	MODERATE	Full QM/MM cascade simulation — surrogate value confirm
P59	✗	30.0	30	LOW	Full coarse-grained MD morphological transition — surro
P61	✗	30.1	30.0	LOW	Full Monte-Carlo population PBPK — surrogate value conf

14g. Class C Translational Deliverables

Principle	Title	Status	Score / Outcome
P21		skipped_deep_validation_	—
P32		skipped_deep_validation_	—
P45		skipped_deep_validation_	—
P55		skipped_deep_validation_	—
P56		skipped_deep_validation_	—

Narratives

P21: Deep validation passed only 12/28 principles. Translational deliverables withheld until a higher-ranked DDS passes deep validation.

P32: Deep validation passed only 12/28 principles. Translational deliverables withheld until a higher-ranked DDS passes deep validation.

P45: Deep validation passed only 12/28 principles. Translational deliverables withheld until a higher-ranked DDS passes deep validation.

P55: Deep validation passed only 12/28 principles. Translational deliverables withheld until a higher-ranked DDS passes deep validation.

P56: Deep validation passed only 12/28 principles. Translational deliverables withheld until a higher-ranked DDS passes deep validation.

v23: structured outlines for Pre-IND, Grant, Patent will be rendered as fully formatted Word/PDF deliverables.

14h. Top-N Fallback Audit Trail

If the Top-1 DDS fails deep validation ($\geq 70\%$ threshold), the orchestrator falls back to Top-2, then Top-3. Each candidate's outcome and the explicit transition reasoning are recorded below.

Rank	DDS Name	Verdict	Pass %	Promoted?
#1	ApoE-LNP	FAILED	42.9%	—
#2	AAV9-vector	FAILED	35.7%	—
#3	RVG29-Liposome-pH	FAILED	42.9%	—

#1 ApoE-LNP — FAILED

Failure reason: Deep validation FAILED: only 12/28 principles validated (42.9%, threshold 70%). Critical failures: P01, P04, P08, P13, P16.

Transition reason: Falling back to rank #2 because deep physics showed insufficient evidence for this DDS.

Failed principles: P01, P04, P08, P13, P16, P30, P31, P33

#2 AAV9-vector — FAILED

Failure reason: Deep validation FAILED: only 10/28 principles validated (35.7%, threshold 70%). Critical failures: P01, P04, P08, P13, P16.

Transition reason: Falling back to rank #3 because deep physics showed insufficient evidence for this DDS.

Failed principles: P01, P04, P08, P13, P16, P18, P30, P31

#3 RVG29-Liposome-pH — FAILED

Failure reason: Deep validation FAILED: only 12/28 principles validated (42.9%, threshold 70%). Critical failures: P04, P08, P13, P16, P24.

Transition reason: No more candidates in Top-3 — reverting to rank #1 and reporting with FAILED verdict. Researcher should reformulate.

Failed principles: P04, P08, P13, P16, P24, P30, P31, P33

15. Executive Decision Framework

Criterion	Status	Evidence
DLVO colloidal stability	FAIL ✗	V_total = -6 kT
BBB penetration >10%	FAIL ✗	DDS BBB = N/A%
Cardiac off-target	PASS ✓	hERG/Nav/Cav QSAR
Clinical trial GO	NO-GO / REFORMULATE	Response 26% AE 0.0%
Sterilization feasible	2 method(s)	Gamma 25Kgy
Supply chain	HIGH	Score 0/100
OVERALL DECISION	CONDITIONAL GO	Proceed to IND-enabling studies

NOTE: All results are computational predictions. Required wet-lab validation: DLS/Zeta (physical stability), HPLC/NMR (drug content), BBB TEER assay, in vivo PK in rodent model.